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THE LEVEL OF HOTEL RESERVATION SYNCHRONIZATION PROCESSES USING INFORMATION TECHNOLOGY TOOLS

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Abstract

The purpose of this paper is to research hotel reservation synchronization processes if the reservation is made throughout a hotel website or if it is booked through the online travel agency (OTA) distribution channels. The subject of the research will certainly be the individual sales channel whose growth directly affects online booking and has a direct impact on the hotel sales revenue growth. The purpose of this paper is also to try to answer the question whether the real-time booking in the hotel sales is at all possible and which information technology tools hotels in Croatia commonly use to connect to different hotel reservation platforms. Empirical research based on the level of hotel synchronization reservation process was carried out on a sample of 130 hotels in the Republic of Croatia belonging to one of the 11 renowned hotel chains. In the processing and analysis of the data descriptive statistics were applied. The results clearly show that 100% real-time synchronization has not yet been achieved when reservations download to the hotel property management system (PMS), including sending a booking confirmation and special wishes listing information of guests. This research offers an overview of the available modern channel management systems offered in today's hotel market and provide a better link between PMS's with the central reservation system, hotel websites and various online distribution channels such as the global distribution system and OTA partners. The limitations of the conducted research can be seen in the spatial restriction as the research was conducted in Croatia and includes hotel chains, so the obtained results do not refer to the complete Croatian hotel industry but to the best practice in the hotel industry in Croatia. Thus, the previous limitations are of no significance as they can represent guidelines and challenges for future scientific research.

Key words: hotel reservation systems, data synchronization, hotel property management systems, hotel website, booking engine, channel managers, OTA

1. INTRODUCTION

The online travel sales worldwide have increased in double digit every years and the online booking market reached the scale of \$189.62 billion USD in 2017 (eMarketer 2017). Hotel industry is developing new ways of accomplishing their missions by leveraging the power of information and applying network-centric concepts (Chawta 2013, 754). New challenge for hotel managers in the 21st century is how to integrate the new, complex and varied services and IT systems into their existing business operations (Law & Jogaratnam 2005, 178).

Developing a new web application to handle online booking from many marketing channels such as OTAs, GDSs, social networks and walk-in to check-in. In the market there are various hotel reservation platforms with integrated interface for different distribution channels and two-way communication for entering and processing reservations with hotel websites in the PMS system. The main categories of information that need to be distributed are: hotel rates (frequency changing, dynamic), hotel availability (frequency changing, dynamic), hotel information such as room types, package types, amenities, location, contact information, meeting space (infrequency changing, static), and rich contents such as photos and videos (some dynamic and some static), which is a category that is growing in importance. According to Forgacs (2010), Guillet & Mohammed (2015), Schütze (2008), dynamic pricing means that a hotel will change its room rates daily or even within a day if up-to-the-minute market information reveals the need for adjustments. It is based on the recognition that the right rate to charge for a room night is what the customer is able and willing to pay.

Effective and high-speed ICT infrastructure and software applications in the hotel industry are crucial for tourism development. Management within tourism companies use ICTs to undertake a range of tasks that enhance the efficiency of employees in the workplace, notably online reservations (ILO 2010, 41). The hotel industry implements the latest in IT to improve productivity, the services offered to clients and to increase customer satisfaction (Werthner & Klein 1999). The use of an information system is indispensable to deal efficiently with the high competitiveness of the market and offer the high quality services demanded by the clients of a hotel (Caro et al. 2003).

Reservation systems are a particularly interesting area of information technology in hospitality. Their first feature is the importance they have for the hotel as the aim of each hotel is to fill accommodation capacities, in order to increase sales and profit. Another important feature of reservation systems is that they connect the hotel to an external environment or to a client (guest or agencies) located in an external system compared to the hotel. Due to these two features, hotel managers are paying great attention to reservation systems as technological progress is the most dynamic one in this field and the new, innovative ways are constantly

looked for to connect guests to hotels in the tourist market. The online technology is the most applicable here. The Internet is getting faster and more complicated. The hoteliers are trying to keep up with the changes. They monitor and analyze market trends and are constantly associating with new partners. There are currently hundreds of portals and agencies, and so many software solutions, and hoteliers choose the ones which produce the most reservations. No matter who and when a new agencies comes to the market, hotels must always be present everywhere.

The subject of this paper is the information related to the processing and confirmation of reservations through hotel websites and OTA portals. The authors researched and followed how the information is entered in Property Management Systems (PMS) in reservation's modules, the ability to send reservation confirmations, as well as the entry of special guest's wishes at hotel PMS. The following systems have been analyzed for communication between: hotel website to PMS (reservations entry in PMS from hotel's own website), hotel website to guest (sending reservation confirmation from hotel's website), hotel website to PMS (entry of special wishes from hotel's website to PMS), OTA portals to PMS (reservation entry from OTA portals to PMS), OTA portals to PMS (sending confirmation from OTA portals to PMS), OTA portals to PMS (entry of special wishes from OTA portals to PMS).

An overview of the current research on the use of ICT in tourism and the hotel industry is given in the second chapter, with special reference to online booking. The third chapter contains an overview of the research methodology and the analysis of the research results on the synchronization of the reservation processes in real time. Finally, the conclusion includes the notes on the role of online booking and the level of synchronization of reservation processes using IT tools. There are also some limitations of this research, and Authors have suggested guidelines to Croatian hoteliers which will enable them to adhere to the world's online reservation trends and compete with the other participants on the online tourism market.

2. LITERATURE BACKGROUND

The development of ICT technology in tourism and the hotel industry and its application has a long history since the first reservation system has emerged and continues in to the future. According to Pease & Rowe 2005, Bilgihan & Bujisic 2015, Law & Jogaratnam 2005, Murphy & Chan 2014, three major waves of technological development have established ICT in hotel companies: (1) the development of Computer Reservation Systems (CRS) in the 1970's; (2) development of Global Distribution Systems (GDS), 1980's and (3) Internet development in the 1990's of the last century. Chronological overview of previous research focused on the development and use of ICT in hotel companies is presented in Table 1.

Table 1. An overview of current research on ICT application in tourism and hotel industry

Authors	Research focus
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Authors	Research focus
Law, R. & Jogaratnam, G. (2005)	A study of hotel information technology applications
Morosan, C. & Jeong, M. (2008)	Users' perceptions of two types of hotel reservation Web sites
Crnojevac, I.H., Gugić, J. & Karlovčan, S. (2010)	eTourism: A comparison of Online and Offline Bookings
Bilgihan, A., Okumus, F., Nusair, K.K. & Kwun, D.J.-W. (2011)	IT applications and competitive advantage in hotel companies
Estis Green, C. & Lomanno, M. (2012)	Distribution Channel Analysis: a Guide for Hotels
Gulmez, M., Ajanovic, E. & Karayun, I. (2014); Seresceanu, R. & Rus, R.V. (2014)	Cloud technology in hotels, cloud based PMS vs. classical PMS,
Anuar, J., Musa, M. & Khalid, K. (2014)	Smartphone's Application Adoption Benefits Using Mobile Hotel Reservation System (MHRS) among 3 to 5-star City hotels in Malaysia
Murphy, H.C., Chen, M.M. (2014)	Online information sources used in hotel bookings
Bemile, R., Achampong, A. & Danquah, E. (2014)	Online Hotel Reservation System
Law, R., Buhalis, D. & Cobanoglu, C. (2014).	Progress of information and communication technology (ICT) in tourism and hospitality industry
Murphy, H.C., Chen, M.-M. & Cossutta, M. (2016)	Multiple devices and information sources in the hotel booking process
Piccoli, G., Lui, T.-W. & Grün, B. (2017)	IT-enabled customer service systems on service personalization, customer service perceptions, hotel information systems, direct and intermediated channels

Source: author's research

The hotel industry is very competitive and customers are become increasingly sophisticated and discerning, demanding high level of quality and value. According to Piccoli et al. (2017) IT-enabled service personalization creates financial benefits for the hotel via revenue share-shift from costly intermediated to direct distribution channels and information system enhances the efficiency of the service personalization process.

The potential tourist surfs through so many websites and promotional messages before making a final decision on which hotel to book such as chain websites, third-party websites, social media channels, search engines, mobile search. (Aggarwal & Aggarwal 2010, 20). It depends on the hotel and its marketing strategies and activities to determine which of those stops will move the guests to generate business in its hotel. According to Murphy & Chen (2016) how the guest will book accommodation presents critical information for hotel marketers who have to decide

where to invest their marketing budget, develop apps and invests in smartphone friendly platforms and strategic partners.

Based on a review of papers published in tourism and hospitality journals between 2009 and 2013, Low et al. (2014) provide an overview of current research in the field of ICTs in hospitality and tourism from the perspectives of consumer and supplier. Content analysis showed that hospitality and tourism industries use ICT in different functional units and for different applications. Of 107 journal papers, there are very few papers dealing with the problem of online hotel booking and its synchronization through rate and revenue management using modern ICT tools.

In raising its competitiveness, the hotel must invest in the development of a modern reservation system. The Central Reservation System (CRS), the reservation system on hotel's websites, the mobile version of the online reservation system must enable customers to book for whatever they need from wherever location they are before lodging into the hotel (Bemile et al. 2014, Anuar et al. 2014).

The CRS Mobile Version automatically identifies if the guest connects to the system via a mobile phone, and adjusts the CRS interface to mobile viewing. US mobile travel sales will total \$75.85 billion in 2017, up 16.7% over 2016 year. On the flip side, desktop and laptop travel sales will decline 1.6% this year to \$113.77 billion. Meanwhile, desktop/laptop's share of all digital travel sales will shrink to 60.0%. And by 2021, mobile will represent nearly half of all digital travel sales (eMarketer 2017).

The smart phone market has also enabled the creation of a smartphone application market that creates value for customers in a number of areas, including the hotel industry (Vigar-Ellis & Bredican 2014). The mobile application allows the guest and hotel staff to interact and offers a variety of hotel facilities using a digital platform. Hotel guests are therefore offered the ability to create a digital mobile key for entering the room and other areas in hotel, opening parking ramps, accessing additional amenities such as saunas and swimming pools, booking excursions, ordering transportation, purchasing items from the hotel's web services etc. At the same time, the hotel can track guest habits through the use of mobile apps, and then monitor the efficiency of the hotel employee's work and with targeted marketing to increase service sales and hotel partner products (Ćosić 2017).

The use of cloud technology has not stalled hotel business either. Cloud based PMS has many advantages over desktop PMS, such as: developing personalized service awareness; its features can be developed or improved easily high security control; more open to the end users (enhanced CRM); online reservations received are directly connected to the PMS and easy follow up on online reservation. Even though keeping up with the modern developments in IT sector is crucial for the success and competitiveness of a hotel, it is usually very hard for new technologies to be accepted and implemented (Gulmez et al. 2013, Sereseanu et al. 2013).

Using Web solutions, the booking sales cycle often includes various intermediaries on the path from the hotel to the customer, resulting in the emergence of the negative aspects of increased technological capabilities. In some cases, the number of intermediaries between hotel room and guest would climb up to five, making the distribution more complex and expensive (Crnojević et al. 2010). Therefore, hotels have to carefully choose distribution channels or combination of

booking channels, and by that not ignoring hotel websites and direct online marketing. Many hotels rather choose reservations through their own website because they primarily save money on agency fees and commissions.

Guests booking through hotel web sites are more loyal, generate more revenue and are in the habit of travelling more often. This is the way how it is supposed to work, but because of OTA's investment in marketing campaigns, hotels are forced to spread their rates and inventory to third party websites to drive volume to their hotel. Morosan & Jeong (2008) evaluate users' perceptions of two different channels for hotel online reservations – hotel websites and third-party websites. Overall, users had more favourable attitudes and higher intentions to revisit OTAs websites than hotel-owned websites.

The Complex Hospitality Reservation Network consists of the major components of the distribution system: Central reservation system (CRS), Property management system (PMS), connectivity to Global distribution system (GDS), Online travel agency (OTA), search engine, Channel management, Branded websites for hotel, Revenue management tools (RMS) and Content management system (CMS). The primary types of connections between distribution partners are from a hotel's CRM to GDS and to OTA. While most use an industry switch/channel partner for this link (such as Pegasus, HBSi or Derbysoft), some include a channel management tool to permit a hotel to update in one place for multiple connections to smaller third party vendors. The switches use their extensive databases to populate many websites, and they related changes from GDS and Internet sites back to the hotel central reservation system (CRS) so that the hotels can service the business (Estin Green & Lomanno 2012, 49).

Some individuals hotels have direct relationships with OTAs by using connections with some manual intervention, such as an extranet, or there is still widespread use of email or fax. Some OTAs also store rates and inventory, and then send a message to the hotel CRS for booking only. Many hotel groups or independents use a third party reservation company, such as iHotelier, Micros-Fidelio or Synxis, to provide most connections, but they may still maintain a few extranets to OTA's for which the updating is often streamlined by the use of a channel management tool, often incorporated into the reservation system (Pilepić et al. 2015).

3. EMPIRICAL RESEARCH

In the following part of the paper research methodology and the results are described.

3.1. Research methodology

In this study survey method was used. The research instrument was a structured questionnaire consisting of four parts as follows: (1) general information and characteristics of the hotel chain, (2) sales channels, (3) online room booking, (4) ICT tools. The first set of questions was related to some of the basic features of

hotel chains, such as: the number of hotels in the company, the category of hotels, the total number of rooms and employees in the hotel chain. The second set of questions related to the occupancy share per individual sales channel (individual, group, congress and allotment), and subsegments of the individual channel (OTA, own web, direct sales, coupons). This set of questions pertains to three consecutive years, 2012, 2013 and 2014, in order to analyze (note) the trends of a certain occurrence through the three-year business period of a hotel. The third part of the questionnaire contained identification questions about online booking and synchronization of the hotel reservation processes. The questions are about how to enter information about processing and booking reservations through hotel websites and OTA portals, sending reservation confirmations, as well as entering special guest wishes. The last part of the questionnaire contains questions related to the type of system used.

The results were analysed by using the software package SPSS (Statistical Package for the Social Sciences). The data were analysed by using the method of descriptive statistics. The research was conducted in the period from March 2015 to January 2016 and included 130 hotels throughout Croatia belonging to one of 11 reputable hotel chains. The research was conducted by Sales Managers, Reservation department or in Revenue Management department. In the sample of 130 hotels, the hotels of the hotel chain Valamar Hotels & Resorts (32.3%) have the largest share, followed by the Maistra Hotels & Resorts hotel chain with 13.8% and Remisens Hotels & Villas with 13.1% (Table 2).

Table 2. Hotel chains covered in empirical research

Hotel chains	Number of hotels	(%)
Arenaturist / Park Plaza Hotels & Resorts	7	5.4
Remisens Hotels & Villas	17	13.1
Crvena Luka Hotel & Resort	1	0.8
Falkensteiner Hotels & Residences	8	6.2
Hilton Hotels & Resorts	2	1.5
Lošinj Hotels & Villas	8	6.2
Aminess Hotels & Camps	4	3.1
Maistra Hotels & Resorts	18	13.8
Laguna Poreč Hotels Resorts & Camping	13	10.0
Starwood Hotels & Resorts Worldwide	6	4.6
Valamar Hotels & Resorts	42	32.3
Vienna International Hotels & Resorts	4	3.1
Total	130	100.0

Source: author's research

Sample consists of hotels of all categories, while the most prominent are 4 stars hotels (49.2%), followed by 3 stars (28.5%) and 2 stars (11.5%) hotels. The smallest share is made up of 5 stars hotels (9.2%).

Table 3. Average number of rooms in a chain by hotel category

Number of hotel rooms	Min	Max	Mean	Std. Deviation
Number of rooms in 5stars	0	655	191.75	201.820
Number of rooms in 4stars	150	2901	831.29	769.461
Number of rooms in 3stars	0	2799	638.23	796.122
Number of rooms in 2stars	0	1424	259.15	424.985

Source: author's research

Hotel chains have on average 831 4 stars category rooms, while category 5 stars on average 192. In addition, hotel chains have on average 638 category 3 stars and 259 category 2 stars.

The subject of the research is the individual sales channel, which has a significant growth in the appearance of the Internet and OTA distribution channels, and which directly influences hotel revenues for higher / higher rates. Individual sales channels are on average occupancy rate of 47.27%, with the lowest occupancy rate being 15% and the highest average occupancy rate of 64%. Looking at the occupancy rate of individual channels in the period from 2012 to 2014, the increase in average occupancy rate from was 31.28% to 38.11%.

Additional benefits of individual channel sales are: functionality (possibility of comparing a larger number of hotels and rates, viewing of guest comments – incentive for booking or not, the possibility of dynamic service creation, greater market reach – visibility on a large number of markets); efficient marketing channel (fast response to market change, distribution support and know-how, result-based collaboration, monitoring and reporting on performance, co-existence with corporate reservation systems) and the cheapest distribution channel (business control is on the side of the hotel because pricing policy, brand identity and direct customer relationship, and thus controlling profitability).

3.2. Research results

The results of the survey on the hotel reservation synchronization process are presented below, according to individual subsegments of the individual sales channel – via the hotel website or through OTA portals (Šimunić 2017).

3.2.1. Contemporary department organization of receiving and processing of reservations – subsegment hotel website

Booking methods will be clearer through the results of the research that follows, and show how to make online reservations from the web site by steps. The

following table shows ways of entering information in the PMS from the website of the hotel.

Table 4. How information is entered in PMS from hotel website and type of system used

Hotel website to PMS	Number of hotels	%
Person	4	3.1
System	120	92.3
Person and system	6	4.6
Total	130	100.0
Name of system		
<i>PHOBS</i>	41	31.5
<i>Trust</i>	8	6.2
<i>My Fidelio</i>	8	6.2
<i>Opera /Valhalla</i>	6	4.6
<i>Sabre / SynXis</i>	54	41.6
No answer	13	10.0
Total	130	100.0

Source: author's research

Table 4 shows that information from the hotel website is most commonly entered by the PMS system (92.3%). Very rarely this is done by only a person or a combined person and system. Information from the web in PMS is most commonly introduced through the PHOBS (31.5%) and Sabre (41.6%) systems.

The PHOBS system is a Croatian product well known outside Croatia. As virtual sales office (VSE) for hotels, resorts, camps and villas, it offers revenue generating solutions for hospitality: property management interfaces, booking engine, yield manager, channel manager, revenue management interface, business intelligence & analytics, B2B manager, another (PHOBS, 2018). What is important is that it integrates a different PMS interface for direct downloading to PMS.

Sabre is one of the world's largest software companies. What started as a joint initiative between American Airlines and IBM to create the world's first computerized airline reservation system in 1960 has since evolved into a technology ecosystem that touches almost every stage of your travel experience (Sabre 2018). Their most famous product SynXis Central Reservations (CR) is the most robust, flexible, easy-to-use rate and inventory management SaaS application. These tools enable hoteliers to support a multitude of rate, inventory and distribution strategies to achieve their business goals. In addition to direct channels, SynXis delivers rates and inventory to all major GDS systems (Sabre, Amadeus and Travelport), as well as online travel agencies through direct connections and switch partners.

Myfidelio.net is a business unit of Oracle Hospitality Ireland Ltd, which is a subsidiary of MICROS Systems Inc. MICROS is the only full systems solutions provider for all market segments with a global distribution and service network in place for major chains, regional chains, local independents, table service, and the

quick service market. FIDELIO Software offering software systems and specializing in the development of Property Management Systems. Myfidelio.net offers Software as a Service business model for hotels and hotel groups as an alternative business option to purchasing licenses. Myfidelio.net is based on OPERA Reservation System (ORS) and hosted in a MICROS datacentre environment in Frankfurt am Main, Germany. It helps generating hotel revenue by enabling consumers to conveniently book hotel reservations online through the most cost effective and successful channel, the hotels' website. It provides real time direct access to the 4 major GDS and over 500.000 travel agencies globally. It links directly to a number of important Internet portals and online travel agencies (Myfidelio.net 2018).

Trust International is now part of Sabre Hospitality Solutions, providing the industry's leading technology for hotels around the world (Trust-Sabre Hospitality 2018).

According to del Rio (2016), a former SVP of Technology Solutions for Starwood, Valhalla is a highly customizable hotel modern Reservation, Call Center, eCommerce Web, Property Management and Revenue Management systems integrated with the Loyalty system via SOA interfaces and open system frameworks.

The following table shows how hotel reservation confirmations are downloaded from hotel website and system types.

Table 5. Ways of reservation confirmation sending from hotel website and name of system used

Hotel website to guest	Number of hotels	%
Person	1	0.8
System	129	99.2
Total	130	100.0
Name of system		
<i>Synxis*</i>	17	13.1
<i>PHOBS</i>	40	30.8
<i>Trust</i>	8	6.2
<i>My Fidelio</i>	8	6.2
<i>Sabre</i>	37	28.5
<i>CRS</i>	6	4.6
<i>WBE**</i>	4	3.1
No answer	10	7.7
Total	130	100.0

** Synxis and Sabre are the same software, Sabre is the company name, Synxis is a program.*

*** WBE – product of the company wbe.travel, web booking expert – a innovative travel booking software providing reliable booking engines to travel agents tour operators and hotel service providers (Web booking expert 2018).*

Source: author's research

In almost all hotels in the sample booking confirmation from hotel website is sent by the system. To send a booking confirmation from the hotel website system PHOBS (30.8%) and Sabre (28.5%) - Synxis (13.1) are the most commonly used.

Table 6. Ways of downloading special wishes from hotel website and name of system used

Hotel website to PMS	Number of hotels	%
Person	14	10.8
System	116	89.2
Total	130	100.0
Name of system		
<i>Synxis*</i>	17	13.1
<i>PHOBS</i>	23	17.7
<i>Trust</i>	8	6.2
<i>My Fidelio</i>	8	6.2
<i>Sabre</i>	37	28.5
<i>MISH**</i>	4	3.1
<i>CRS/Valhalla</i>	6	4.6
No answer	27	20.8
Total	130	100.0

* *Synxis and Sabre are the same software, Sabre is the company name, Synxis is a program.*

** *MISH – the product of ISTRATECH Pula, Hotel System and Catering Modular Information System, a software package for the entire business of hotel and catering companies, representing ERP (Enterprise Resource Planning) solution (Istra Tech 2015).*

Source: author's research

In almost 90% of hotels in the sample, the special wishes of guests with hotel websites are entered through the system. Hotels have well-developed reservation systems but still all guest wishes entered the hotel website have to be processed further by processing all comments in online reservations within 48 hours, depending on the day of arrival.

In the case of selling extra services / costs (upgrade, extra beds, additional services), and if the guest confirms it – the person (hotel reservation agent must in any case enter the information in the PMS through the note in the reservation. The special wishes of the guests from the hotel website are most often downloaded through the Sabre system (28.5%) – Synxis (13.1) and PHOBS (17.7%).

3.2.2. Contemporary department organization of receiving and processing of reservations – subsegment OTA portals

This subsection shows the method of distributing reservations and processing them through various OTA portals that hotels use.

Table 7. Ways of downloading information from OTA portals and name of system used

OTA portals to PMS	Number of hotels	%
Person	4	3.1
System	126	96.9
Total	130	100.0
Name of system		
<i>Synxis-XML-MISH*</i>	17	13.1
<i>PHOBS</i>	41	31.5
<i>Trust</i>	8	6.2
<i>My Fidelio</i>	8	6.2
<i>Valhalla</i>	6	4.6
<i>Sabre</i>	37	28.5
No answer	13	10.0
Total	130	100.00

* *Synxis-XML-MISH* –direct two-way XML connections between Synxis software and the PMS. This provides constant updates and availability to the OTAs.

Source: author's research

Information from OTA portals are most commonly downloaded through the system (96.9%) in PMS. The most commonly used systems are PHOBS (31.5%) and Sabre (28.5%).

Table 8. Ways of reservation confirmation sending from OTA portals and name of system used

OTA portals to PMS	Number of hotels	%
Person	13	10.0
System	117	90.0
Total	130	100.00
Name of system		
<i>OTA</i>	95	73.1
<i>PHOBS</i>	5	3.8
<i>Trust</i>	8	6.2
No answer	22	16.9
Total	130	100.0

Source: author's research

The results in Table 8 show that 90% of the hotels use the system to send a reservation confirmation from OTA portals. The OTA's software (73.1%) is most often used to send a reservation confirmation from OTA portals.

Table 9. Ways of downloading special wishes from hotel OTA portals and name of system used

OTA portals to PMS	Number of hotels	%
Person	66	50.8
System	64	49.2
Total	130	100.0
Name of system		
<i>Synxis-XML-MISH</i>	17	13.1
<i>PHOBS</i>	28	21.5
<i>Trust</i>	8	6.2
<i>Valhalla</i>	2	1.5
No answer	75	57.7
Total	130	100.0

Source: author's research

The practice of entering/downloading special wishes of guests from OTA portals is similar. Approximately 50% of the hotel uses the system, and the rest is performed by the person. For downloading special wishes from OTA portals, most commonly used are PHOBS (21.5%) and Synxis-XML-MISH (13.1%).

4. DISCUSSION AND CONCLUSION

Today's market and hotel software solutions development as a as well as web portals participating in the sales process of hotels or hotel chains completely determine the concept of rate distribution and sales strategies, especially marketing activities and sales promotion of services offered. One of the greatest challenges for hotels is to regularly set up the current rates on booking portals and to update the reservation status in real time. The goal is to reduce manual entry and have up-to-date rates, accurate room occupancy levels – ARI (availability, rates and inventory) (Ling et al., 2015). If the ARI connection is not well set up and synchronized to the maximum as current technology allows, there is a risk of overbooking as well as rate parity policy violations.

This research details the steps required for online reservations, booking confirmation, as well as the special wishes of guests entry either through the hotel website or OTA that reflect the level of synchronization of the reservation process of today's hoteliers. Today, every serious hotel company has its own reservation center, which is part of centralized sales, which is responsible for individual sales channel, especially for its subsegment – Online. As far as the online segment is concerned, it is the largest subsegment of the individual sales channel. Hotel reservation agents are responsible for receiving and processing online (OTA) and web reservations and requests, receiving and processing coupon reservations and requests, and placing and contracting coupon bids. In addition to the above mentioned responsibilities,

reservation centers provide support to guests in providing information about facilities, services, and offers.

Reservation center deals with OTA booking entry in PMS, checking the accuracy of reservations entered through the interface, and processing reservations in terms of special conditions or answering comments – guest requests. Agent manually enters the PMS reservation of those non-WML-related portals to PMS. Reservations are entered directly into the PMS according to the parameters of the OTA confirmation received. If the OTA portals and the hotel website are interfaced with the PMS, reservation is automatically downloaded to the PMS via the interface, and such booking agents do not enter. Modifications are made directly to the PMS when portals are not interfaced with the PMS.

In hotels in Croatia the information from the hotel website in PMS is most commonly entered through the PHOBS and Sabre systems. In almost all hotels in the sample booking confirmation from the hotel website is sent by the system (PHOBS and Sabre). Guest special wishes entered in hotel website are almost completely entered through the system. To send a reservation confirmation from OTA portals, the system is most often used. The practice of entering/downloading special wishes of guests from OTA portals is similar. Approximately 50% of the hotel uses the system, and the rest is performed by the person. It can be concluded that 100% synchronization of the booking process in real time is not yet reached when downloading the reservation in the PMS, sending a confirmation of the booked reservation and entering special wishes.

Although research has been carried out in detail, limitations of the conducted research may be reflected in the fact that research has been carried out at hotel chains in Croatia, and therefore research has a spatial limitation, since it has been implemented uniquely on one country example and no comparison with other, similar or different destinations in the world. Furthermore, the research carried out only covers hotel chains, and the results obtained do not refer to the complete Croatian hotel industry but only to the best business practice in hotel industry in Croatia because the results are based on the characteristic ICT equipment of hotel chains in Croatia.

It should be noted that hotel software takes over people's work due to automatization of processes, and hotel business continues to struggle, since hospitality in hotel business is the most important thing that no software can provide.

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